Recreating Fred Brooks's Office in Virtual Reality Using Photogrammetry

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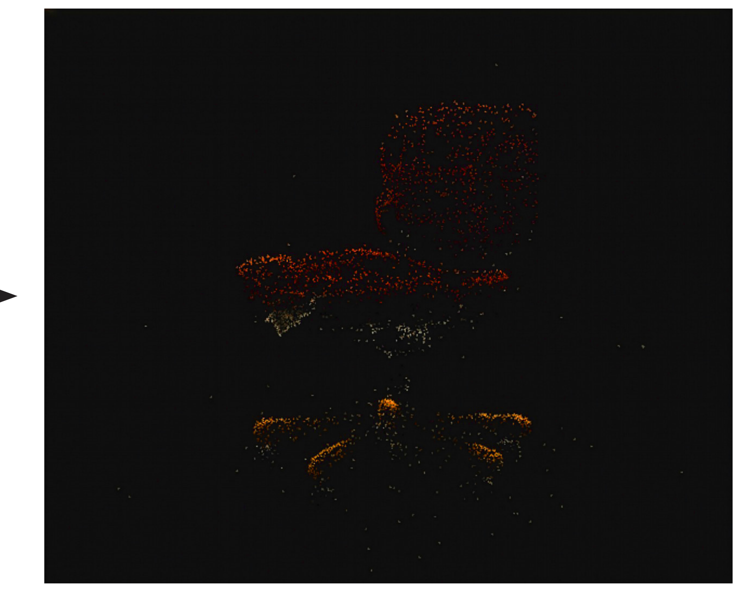
Abstract

This project involved the digital recreation of the late Fred Brooks's office in virtual reality. In order to maximize realism, the models were created using photogrammetry software, which creates 3D models from collections of pictures. A consequence of using this technique is that the resulting 3D meshes are often complex and depending on how the photos were taken, can have partially inaccurate geometry.

To simplify the models for use in virtual reality, we implemented the following workflow: The models of the office and its furniture were created using RealityScan, a photogrammetry software, and were then retopologized using Instant Meshes, an open-source autoretopologization software. Afterwards, the models were placed in Blender, where we simplified them by dissolving faces and vertices. Finally, the simplified models were exported to Unity, where they were implemented in a virtual reality environment. This workflow allowed us to significantly reduce the complexity of the models with minimal deviation from the original shape. The approach demonstrates a practical pipeline for creating 3D models from photogrammetry and transforming them into models suitable for use in VR.

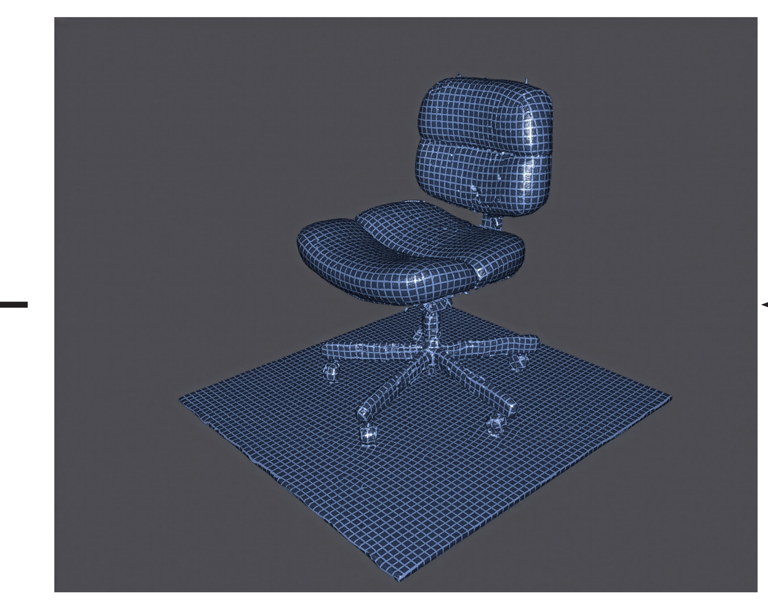
Model Creation Pipeline

Taking Photos to be used in photogrammetry software



Using RealityScan to create point cloud from pictures

Creating 3D mesh from point cloud in RealityScan



Retopologizing 3D mesh in Instant Meshes.

Simplify the retopologized model in Blender

Place simplified model into Unity Project with office model

Furniture models in various stages of the modeling pipeline

Vertex Counts of Models After Each Stage

Model	Base Photogrammetry	After Retopologization	After Simplification	% Overall Vertex Reduction
Office	2,790,694	644,345	197,219	-92.93%
Chair	647,997	47,071	39,794	-93.86%
Desk	1,813,919	39,727	1,678	-99.91%
Printer Table	430,176	xx	32,273	-92.50%
Printer	180,921	xx	422	-99.77%
Phone Table	197,057	45,810	21,537	-89.07%
Radio & Lamp Table	510,663	76,025	15,840	-96.90%
Couch	1,126,482	52,763	44,335	-96.06%

Future Work

While this project was successful in demonstrating a practical pipeline for modeling physical objects in VR, there are several ways in which this workflow could be improved. To address the great complexity of the 3D meshes that Reality Scan outputs, automated simplification methods will be investigated, such as Blender Python workflows, which would allow polygon counts to be decreased with minimal manual intervention. This will also be accomplished with much reduced deviation from the original geometry of the objects.

Dynamic reduction of mesh complexity based on player proximity could also be implemented within Unity to improve performance. For earlier stages in the pipeline a more standardized photography protocol will be developed with specific angles, overlaps, and distances to improve the consistency of mesh reconstruction. More controlled lighting setups and video-based capturing methods where frames are extracted from a slow pan of the object, will also be evaluated as alternatives to individual photographs. By improving capture quality and then connecting that to the automated simplification stage, the pipeline will be made much more repeatable, which reduces the likelihood of errors throughout the workflow.